ECON3700
ENVIRONMENTAL ECONOMICS

Environmental Analysis I

Frameworks of Analysis;

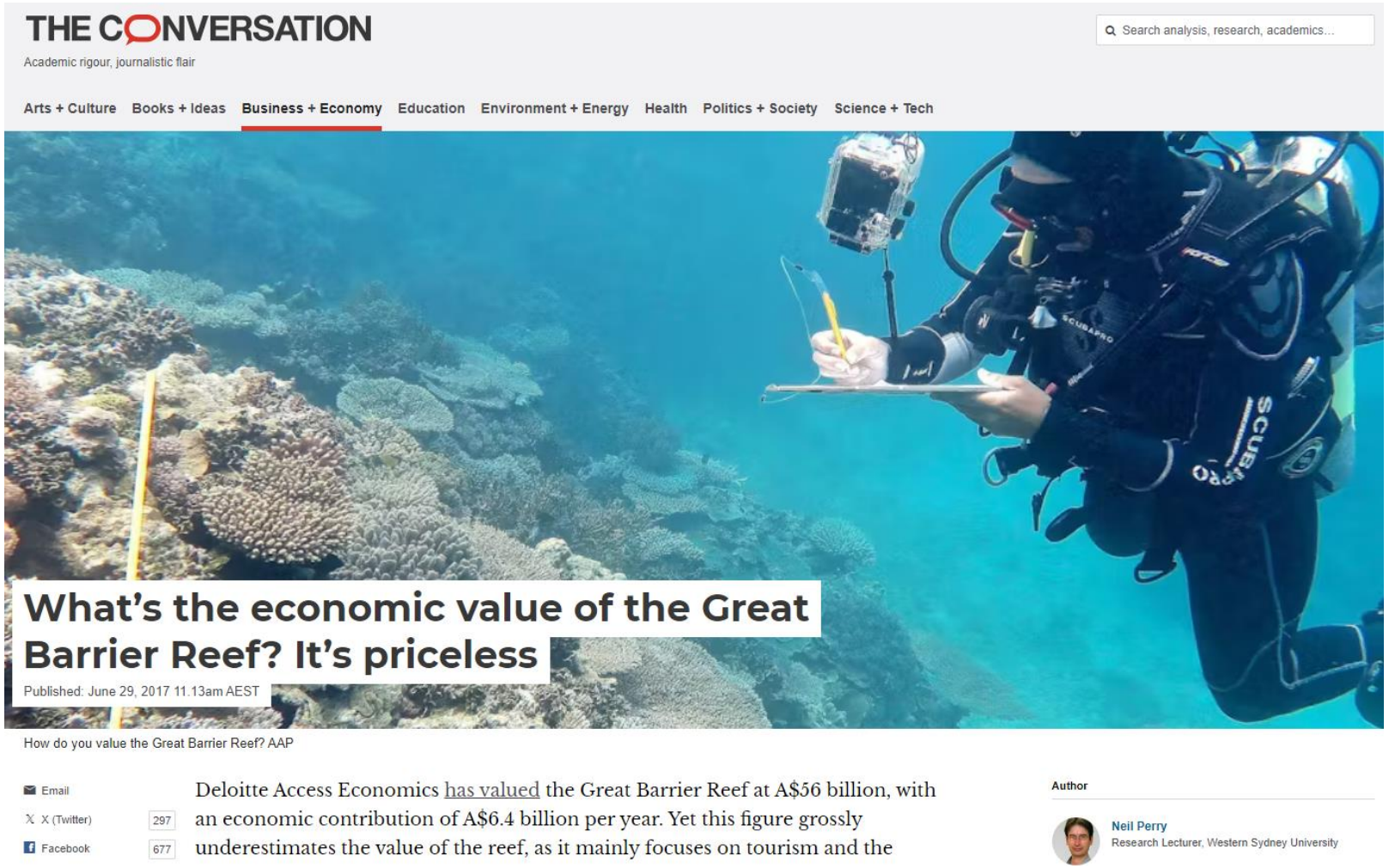
Benefit-Cost Analysis - Benefits

References: Chapter 6 & 7, ISE Environmental Economics, 9th Edition

Presenter: Dr Ammarr Aripin, School of Economics.

Learning Objectives

1. Explain impact analysis and distinguish between environmental, economic, and regulatory impacts.
2. Describe the purpose and application of cost-effectiveness analysis and damage assessment.
3. Define cost-benefit analysis (CBA), outline its process, and differentiate it from efficiency analysis.
4. Apply discounting and present value concepts to compare costs and benefits over time and discuss appropriate discount rates.
5. Assess risk in CBA using expected value calculations and understand different risk preferences.
6. Evaluate equity considerations in project assessment, including horizontal and vertical equity.
7. Identify and distinguish between use values and non-use values of environmental goods.
8. Apply environmental valuation methods, including revealed preference and stated preference techniques.



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What's the economic value of the Great Barrier Reef? It's priceless

Published: June 29, 2017 11.13am AEST

How do you value the Great Barrier Reef? AAP

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Deloitte Access Economics has valued the Great Barrier Reef at A$56 billion, with an economic contribution of A$6.4 billion per year. Yet this figure grossly underestimates the value of the reef, as it mainly focuses on tourism and the

Author

Neil Perry
Research Lecturer, Western Sydney University

Source: [What's the economic value of the Great Barrier Reef? It's priceless (theconversation.com)]

Part A: Impact Analysis

What is Impact Analysis?

Impact means the effects of any actual or proposed policy.

There are many types of effects;

I. Environmental; *Environmental Impact Analysis EIA*
II. Economic; *Economic Impact Analysis*
III. Regulatory; *Regulatory Impact Analysis RIA*

What is Impact Analysis? - Australia

Impact means the effects of any actual or proposed policy.

There are many types of effects;

I. Environmental; *Environmental Impact Analysis EIA*

- *Required under state laws and the EPBC Act (1999)*
- *Assesses impacts on biodiversity, ecosystems, water, Indigenous heritage*
- *Includes public consultation*

What is Impact Analysis? - Australia

Impact means the effects of any actual or proposed policy.

There are many types of effects;

II. Economic; *Economic Impact Analysis*

- *Used by Infrastructure Australia and governments*
- *Evaluates jobs, productivity, regional growth*
- *Often includes Cost–Benefit Analysis (CBA)*

What is Impact Analysis? - Australia

Impact means the effects of any actual or proposed policy.

There are many types of effects;

III. Regulatory; *Regulatory Impact Analysis RIA*

- *Required by the Office of Impact Analysis (OIA)*
- *Ensures regulation is evidence-based and cost-effective*
- *Assesses compliance costs, competition, broader policy alignment*

Home | The Office of Impact Analysis

Part B: Cost-Effectiveness Analysis

Cost Effectiveness Analysis CEA

Compares the costs of different interventions to their outcomes, expressed in non-monetary units, such as health improvements, environmental impact reductions, or energy savings.

How to measure cost-effectiveness analysis?

Cost Effectiveness Analysis CEA

How to measure cost-effectiveness analysis?

1. Define the Objective

- **What outcome** are you trying to achieve?
  Example: Reducing carbon emissions or improving wetland biodiversity.

2. Identify the Alternatives

- List the **different options** or interventions to be compared.
  Example: A new public transport system vs. a bike-sharing program.

Cost Effectiveness Analysis CEA

How to measure cost-effectiveness analysis?

3. Determine Costs for Each Option

- Include all relevant costs (direct, indirect, fixed, and variable costs):
  - Capital investment (e.g., infrastructure)
  - Operational costs (e.g., maintenance, staff wages)
  - Externalities (e.g., environmental damage or benefits)

Express all costs in a common currency, adjusted for inflation and discounted over the period - PV.

Cost Effectiveness Analysis CEA

How to measure cost-effectiveness analysis?

4. Measure Effectiveness (Non-Monetary Outcome)

- Identify the effectiveness metric (often expressed as a single outcome measure):
  - Environmental CEA: Reduction in CO_2 emissions (tons/year)
  - Health CEA: Life-years gained, or cases prevented
  - Education CEA: Literacy rate improvement or student graduation rates

Cost Effectiveness Analysis CEA

How to measure cost-effectiveness analysis?

5. Calculate the Cost-Effectiveness Ratio (CER)

$$CER = Cost / Effectiveness$$

This ratio shows the **cost per unit of outcome** achieved.

Example: If a wetland restoration project costs **$500,000** and reduces CO_2 emissions by **10,000 tons**, the CER is: **$500,000 / 10,000 tons = $50 /ton of CO2**

Cost Effectiveness Analysis CEA

How to measure cost-effectiveness analysis?

6. Compare Across Alternatives

- **Lower CER is better**: It indicates the intervention achieves the desired outcome at a lower cost.

Example:

  - **Bike-sharing program**: 30 AUD/ton of CO_2
  - **Public transport upgrade**: 50 AUD/ton of CO_2
  - Choose the bike-sharing program for better cost-effectiveness.

What is Cost-Effectiveness Analysis ?

Cost-effectiveness analysis involves estimating the costs of various options that share a common objective.

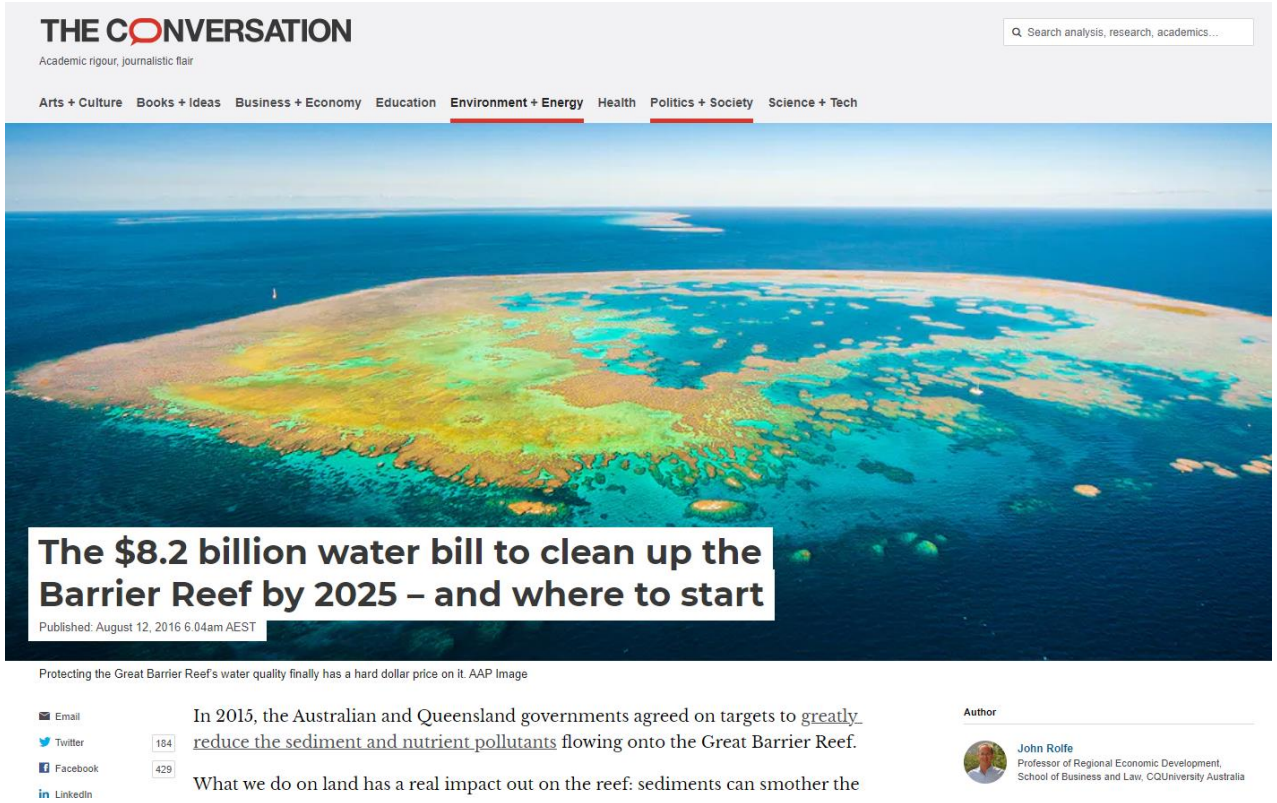
Most cost-effective *vs* **least** cost-effective

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TABLE 6.1 Cost-Effectiveness of Alternative Means of Reducing Methane Leaks in Natural Gas Operations

Technique	Costs per Thousand Cubic Feet
Upgrading pneumatic control devices	1.99
Upgrading compressor pumps	6.89
Installing new systems for liquid extraction	5.03
Replacing pneumatic with solar pumps	4.86
Change venting procedures	0.41
Leak detection and repair at wells	7.60
Leak detection and repair in transmission	2.15

Source: ICF International, *Economic Analysis of Methane Emission Reduction Opportunities in the U.S. Onshore Oil and Natural Gas Industries,* Proposal for Environmental Defense Fund, 2014.



Source: [The $8.2 billion water bill to clean up the Barrier Reef by 2025 – and where to start (theconversation.com)]

Part C: Damage Assessment

What is Damage Assessment?

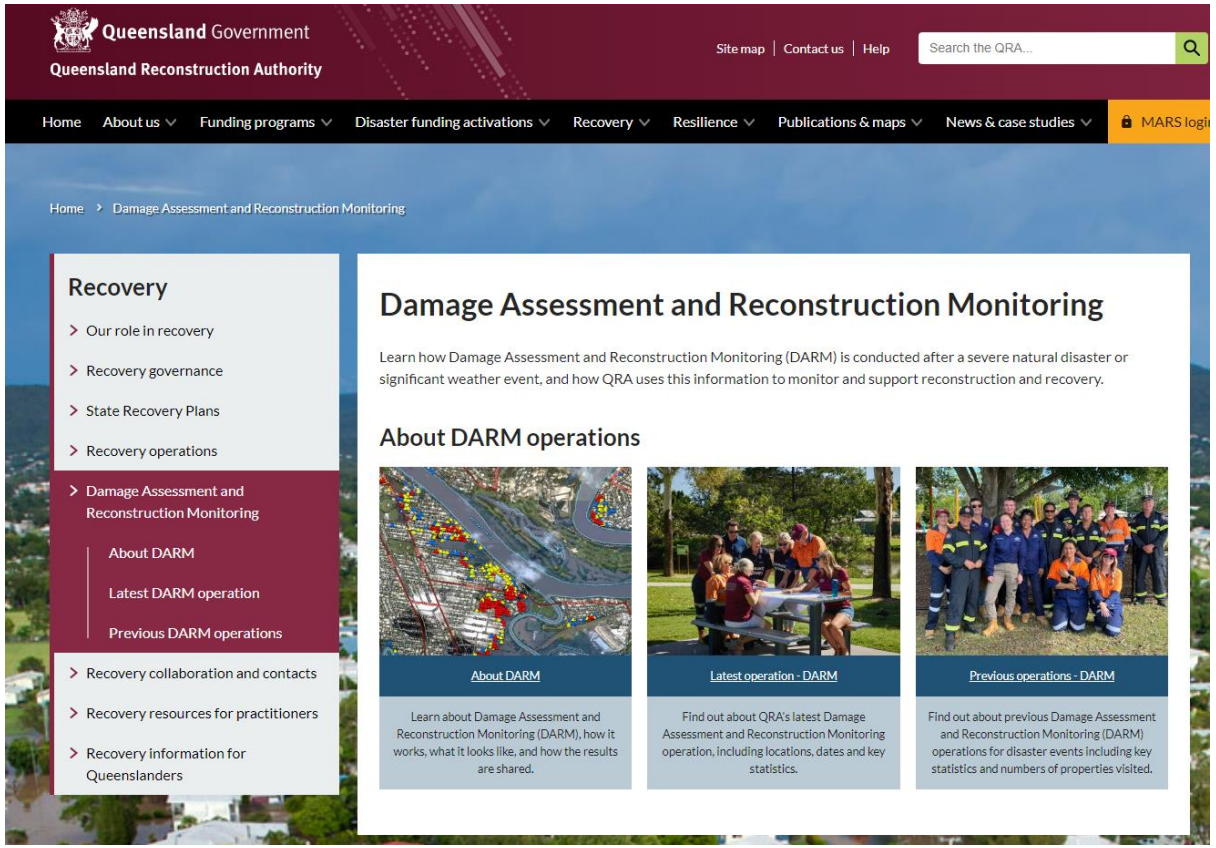
The process of evaluating and documenting the extent of damage caused by natural disasters *(e.g., bushfires, floods, storms etc.)*

Data; destroyed homes, businesses, infrastructure, agriculture and natural resources.

Also estimating the economic losses incurred due to the disaster.

Carried out by various government agencies, emergency services, and state and federal governments.

What is Damage Assessment?



Source: [Damage Assessment and Reconstruction Monitoring | Queensland Reconstruction Authority (qra.qld.gov.au)]

Part D: Benefit-Cost Analysis

What is Cost-Benefit Analysis?

Cost-benefit analysis is:

- a tool to assist decision-makers
- used by public economists to assess new projects and regulations
- based on the idea that for any new intervention (e.g., new road, new vehicle standard) the benefits must outweigh the costs

What isn't Cost-Benefit Analysis?

Cost-benefit analysis is not:

- a tool that captures all factors relevant to a decision
- immune to manipulation

Basic CBA Process

- Define the project/regulatory program explicitly.
- Outline the inputs and outputs.
- Estimate the social costs and benefits of inputs and outputs.
- Compare the benefits to the costs (e.g., benefit-cost ratio).

If the benefits outweigh the costs, is the program necessarily efficient?

CBA vs Efficiency

- We want the project with the highest aggregate net benefits, not any project with positive net benefits.
- Ideally, we can evaluate all alternatives and scales so we can pick the project such that MB=MC.
- In reality we will only be able to evaluate a handful of options.
- We should pick the project with the highest benefit-cost ratio.

CBA of Coal Combustion Residue Regulations

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	$ Millions, 2018 prices 7% discount rate, 50-year life	
Benefits:		
Value of human cancer risks avoided	44	
Groundwater remediation costs avoided	40	
Cleanup costs avoided	797	
Induced future CCR beneficial use*	7,284	
Total		8,166
Costs:		
Engineering controls	584	
Ancillary regulatory requirements	127	
Conversion to dry CCR disposal	1,042	
Total		1,754
Net Benefits		6,412

*The assumption is that by increasing the cost of handling CCR, firms would be more creative in finding markets for CCR for beneficial use by other industries.

Source: U.S. EPA, Office of Resource Conservation and Recovery, *Regulatory Impact Analysis for EPA's Proposal RCCA Regulation of Coal Combustion Residues (CCR) Generated by the Electric Utility Industry,* 2010; updated to 2018 Bureau of Labor Statistics prices.

Time preferences

Costs and benefits will often occur at different points in time.

People generally prefer to get a dollar today compared to a dollar tomorrow. This means people like prefer:

- to bring forward benefits
- to delay costs (pay tomorrow rather than today)

So, when costs and benefits occur is important to the value of a program.

We need a way of adjusting costs and benefits to reflect when they occur.

Discount rates

Say that the simple interest rate for one year is 20%.

If I invest $100 today, how much would I receive:

- a year from now? **$120**
- two years from now? **$144**

Discount rates

- simple interest rate r
- today's investment y

How much money will I have in my bank account:

- In one year? **(1 + r) y**
- In 2 years? **(1 + r)2 y**
- In T years? **(1 + r)T y**

If I want to have $200 in the bank next year, how much do I need to invest now?

How much would I need to invest today to have $200 in the bank in 2 years?

Present Value

- how much do I need to invest today to have x in the bank in T years
- value I get today from payments that take place in the future

$$PV = \frac{x}{(1+r)^T}$$

Present values allow us to compare costs and benefits that occur at different points in time in the same units.

Time of Costs and Benefit Matters

A lot of environmental problems, e.g., climate change:

- incur costs now
- benefits many years into the future
- discounting is critical

Understanding the Implications of Discounting

What is the present value of $1 million in 2050?

Discount rate	Present value
1.4%	$499,000
6%	$54,000
10%	$9,000

At a 10% discount rate, willing to pay no more than $9,000 today to avoid $1 million in losses in 2050.

What is the right discount rate?

- Subject of much debate
- Some people argue that the discount rate should be the market rate: markets efficiently allocate resources over time (weak sustainability)
- Some people argue that for environmental projects should be zero (strong sustainability)
- Common ranges used in environmental projects: 1.4% (Stern) - 6% (Nordhaus)

Part E: CBA, Risk and Equity

Risk

Neither physical nor economic outcomes are known with certainty.

There are always risks.

How should we factor that risk into our CBA?

Expected Value

We can compute the expected value if we know the probability of different outcomes.

Example

- The value of the good outcome is $100
- The value of the bad outcome is $0
- The good outcome occurs with a probability 0.4 (40% of the time)
- The bad outcome occurs with a probability 0.6

Expected Value

EV = [Probability good]* [Value good]
+ [Probability bad]* [Value bad]

$$EV = [0.4] * [\$100] + [0.6] * [\$0]$$

$$EV = \$40$$

Risk Preferences

Risk neutral:

Indifferent between $1 for certain and a gamble with $1 EV

Risk averse:

Prefer $1 for certain to a gamble with $1 EV

Risk loving:

Prefer gamble with $1 EV to $1 for certain

Risk Preferences

If the decision maker is risk neutral:

- evaluate the EV of costs and benefits

If the decision maker is risk averse:

- prefer projects with the same EV of costs and benefits but less risk

Risk Preferences

Program A		Program B	
Net Benefits	Probability	Net Benefits	Probability
$500,000	0.475	$500,001	0.99
$300,000	0.525	- $10,000,000	0.01
Expected value	$395,000	Expected value	$395,000.99

Which program would a risk neutral decision maker prefer?

Strictly prefer Program B

Which program would a risk averse decision maker prefer?

Likely to prefer Program A

Distribution and Equity

CBA is a tool to compare costs and benefits.

It does not take into account of other concerns like distribution and equity

(NB: neither does the concept of efficiency!).

Distribution and Equity

Concepts of equity:

Horizontal equity: compare how people in similar socio-economic circumstances are treated
e.g., middle-income earners in the city vs in the country

Vertical equity: compare how people in different socio-economic circumstances are treated.
Examples

- Proportional impact: the costs and benefits are distributed in proportion to income
- Regressive impact: higher proportional net benefits to high-income people
- Progressive impact: higher proportional net benefits to low-income people

Part F: Quantifying Environmental Benefits

Quantifying Environmental Benefits

Because of market failure, environmental goods and services do not have prices.

How do we value environmental benefits?

Use Value

Benefits people get from direct use of a good

- eating fish
- visiting the Reef
- better health

Non-Use Value

- Existence value: WTP to help preserve the existence of an environmental amenity (endangered coral)
- Option value: WTP to preserve the option of being able to experience a particular environmental amenity in the future (visit the Reef someday)
- Bequest value: WTP to leave behind environmental quality for future generations
- Stewardship value: WTP to preserve the environment not just for human use, but for all living organisms

How to Measure Environmental Benefits?

Revealed preference: infer the value of environmental goods from other market transactions (use values)

- Defensive expenditures
- Hedonic analysis
- Travel cost

Stated preference: ask individuals, about hypothetical situations (use and non-use values)

- Contingent valuation

Revealed preference: Defensive Expenditure

Examples:

- Bottled drinking water
- Air filters
- Diesel generator
- Fences to hide polluted site

Spend as long as marginal benefits of reduction > marginal cost

Defensive expenditures = lower bound on MWTP for environmental good; lower bound on benefits individuals would receive if harm were removed

Revealed preference: Hedonic Analysis

- Premise: price of a marketed good is function of its characteristics
- Value individual characteristics by looking at how the price people are willing to pay for the good changes when characteristics change
- Data: market supply-demand equilibria

Examples:

- variation in housing prices reflecting value of local environmental attributes, like air quality or proximity to toxic facility
- variation in land prices reflecting expected profitability of agricultural production f(climate)

Revealed preference: Travel Cost

Intuition: visiting a site takes time and money

Travel cost approach: use price-quantity variation in travel cost to trace out WTP

- price: distance to park, the opportunity cost of time, expenditures getting to/at site
- quantity: number of visitors

Travel cost = lower bound on benefits individuals receive from a site

Components of Travel Cost

Direct costs:

- airfare
- entrance fees
- camping, fishing equipment

Indirect costs:

- opportunity cost of travel time